Ans 1.

The OSI model has 7 layers:

1. Physical Layer: Transmits raw bits over cables / media.
2. Data Link Layer: Handles node-to-node transfer & MAC framing.
3. Network Layer: Manages IP routing and packet forwarding.
4. Transport Layer: End-to-end delivery & error checking (e.g. TCP).
5. Session Layer: Manages sessions between applications.
6. Presentation Layer: Syntax, encryption & data formatting.
7. Application Layer: User interface & protocols like HTTP.

Ans 2.

Given matrix $A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

Step 1: Find Determinant

$$\det(A) = (4 \times 6) - (7 \times 2) = 24 - 14 = 10$$

Since $\det(A) = 10 \neq 0$, inverse exists.

Step 2: Adjoint of matrix

$$\text{adj}(A) = \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

Ans 3.

Ans 4.

Binary Search;

Precondition: array must be sorted.

Working: low = 0, high = n-1.

mid = (low + high)/2

if arr[mid] == key → return mid

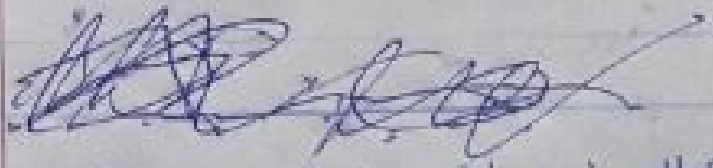
if key < arr[mid] → high = mid-1

else → low = mid+1

Repeat while low <= high.

Time complexity = O(log n).

Ans 5.



Light rxn takes plee in thylkd mmbrne...

chlrofl absobs phtns...

wtr splts $2H_2O \rightarrow$ ...

elctr trnsprt chn mkng ATP & NDPH...

rleses Oxgn.

